Selection

CSE 101 Section 03
Introduction to Algorithms and Computing
Fall 2023
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Selection

If you are in the kitchen:

get a snack

If you are taking CSE 101 and MAE 101

    stand on your head

else

    do a cartwheel

```
If you are wearing green or orange
        sing a song
else if you are wearing blue
        chirp like a bird
else if you are wearing pants
        bark like a dog
else
        quack like a duck
```

The if statement

- Matlab syntax:

```
if condition:
    statement
    statement
```

- First line known as the `if` clause
  - Includes the keyword `if` followed by condition
    - The condition can be `true` or `false`
    - When the `if` statement executes, the condition is tested, and if it is true the block of statements is executed. otherwise, the block of statements is skipped

If condition ?

- The condition in the if statement is a *Boolean expression*, i.e. expression tested to determine if it is true or false
  - Example: a > b
    - `true` if a is greater than b; `false` otherwise

- Conditions rely on *relational operators* (covered in previous lecture) to determines whether a specific relationship exists between two values.

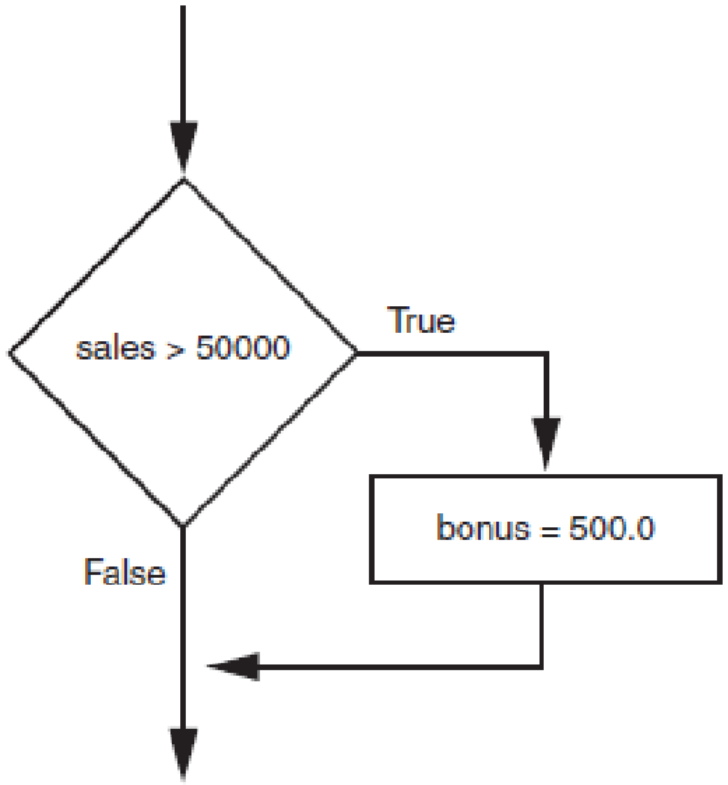
RECALL: Operations: Relational

- Relational operators compare the elements in two arrays and return logical `true` or `false` values to indicate where the relation holds.

==	Determine equality	3 == 2
>=	Determine greater than or equal to	2 >= 1.5
>	Determine greater than	2 > 2
<=	Determine less than or equal to	5 <= 7
<	Determine less than	3 < 3
~=	Determine inequality	3 ~= 3

Example

```
sales = 10000
if sales > 50000:
    bonus = 500
```



Example of testing equality

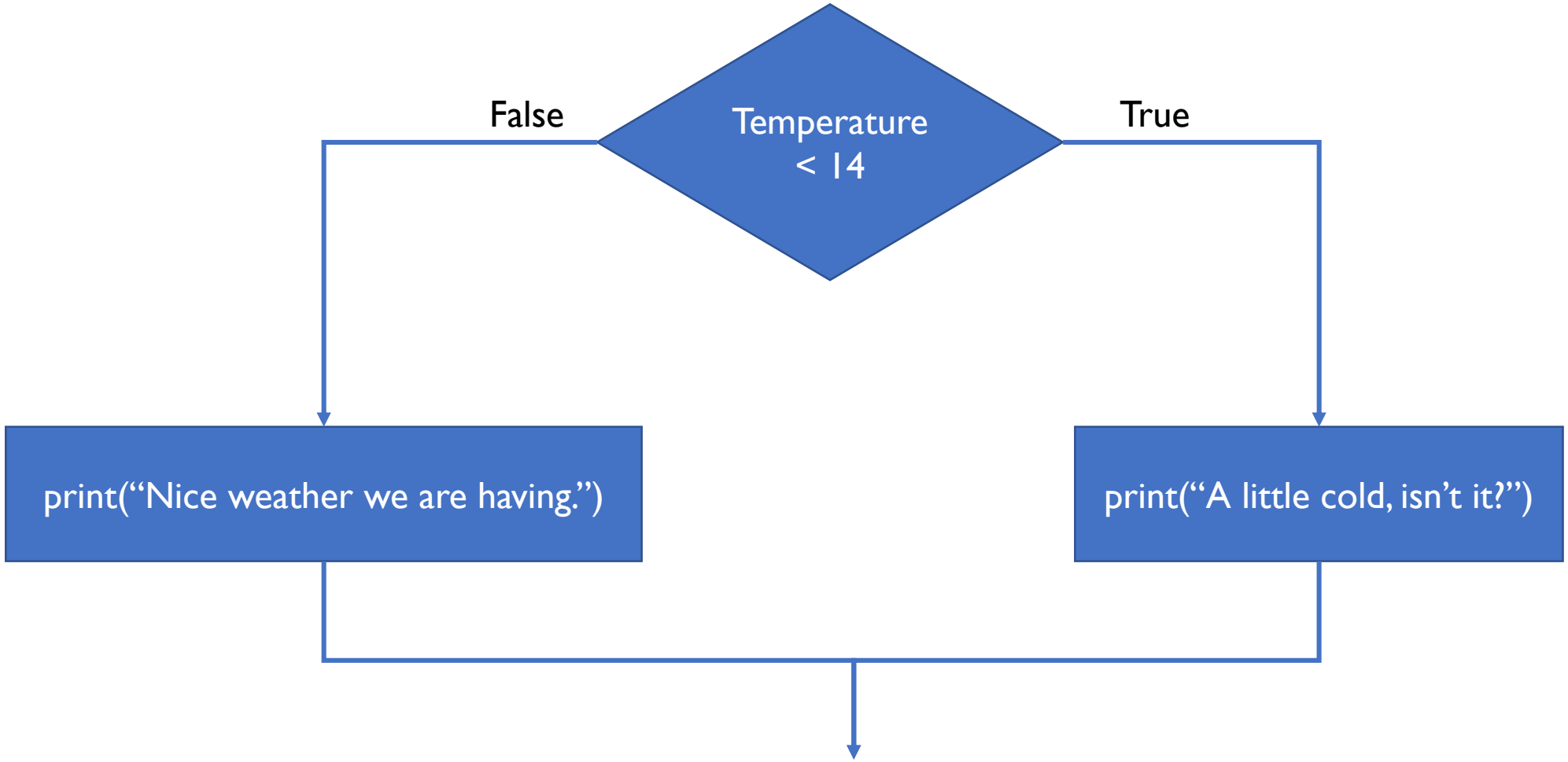
```
if course == "CSE101":
    mark = 100
```

The if-else Statement

- Dual alternative decision structure: two possible paths of execution
  - One is taken if the condition is true, and the other if the condition is false
  - Syntax:

```
if condition:
    statements
else:
    statements
```

If-else example



Need to evaluate more conditions?

- Syntax:

```
if condition:
        statements
elif condition:
        statements
elif condition:
        statements
else:
        statements
```

Example

```
x = 10
minVal = 2
maxVal = 6

if minVal <= x  and x <= maxVal:
    print('Value within specified range.')
elif x > maxVal:
    print('Value exceeds maximum value.')
else:
    print('Value is below minimum value.')
```

Practice 1

Write Python code to:

- Determine if someone qualifies for a loan, they must meet one condition:
  - Must earn at least $50,000/year

Practice 2

Modify Python code so that:

- For someone to qualify for a loan, they must meet <u>two</u> conditions:
  - Must earn at least $50,000/year
  - Must have been employed for at least two years

- *Hint:* Check first condition, and if it is true, check second condition

Building more complex conditions using logical operators

Logical operators: operators that can be used to create complex Boolean expressions

- *and* operator and *or* operator: binary operators, connect two Boolean expressions into a compound Boolean expression
- *not* operator: unary operator, reverses the truth of its Boolean operand

OPERATOR	DESCRIPTION	SYNTAX
and	Logical AND: True if both the operands are true	x and y
or	Logical OR: True if either of the operands is true	x or y
not	Logical NOT: True if operand is false	not x
Exclusive or (xor)	Logical XOR: True if <u>only one</u> of the operands is true	x ^ y

Practice 3

- Modify the code from practice 1 to now use the logical operator *and*.

Truth tables

and

a	b	a and b
0	0	0
0	1	0
1	0	0
1	1	1

or

a	b	a or b
0	0	0
0	1	1
1	0	1
1	1	1

xor ^

a	b	a ^ b
0	0	0
0	1	1
1	0	1
1	1	0

not

a	not a
0	1
0	1
1	0
1	0

Revisiting precedence of operators

Arithmetic operators take precedence over logical operators.

Python will always evaluate the arithmetic operators first (** is highest, then multiplication/division, then addition/subtraction).

Next comes the relational operators.

Finally, the logical operators are done last.

Practice

Evaluate the following Boolean expressions (i.e. are they True or False?)

```
age >= 18 and has_id_card
is_weekend or is_holiday
not is_raining
(x > 0) and ((y < 10) or (z == 20))    # given x = 2, y = 11, and z = 20
(a + 1 < b) ^ (a <= b) # given a = 1, and b = 3
```

Next…

Loops